Using HSI Color Space to Improve the Multispectral Lidar Classification Error Caused by Measurement Geometry

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***Abstract*— Multispectral lidar has become a promising technology with the rise in capability of 3-D spectral imaging. However, the precise acquisition of spectral information is interfered by measurement geometry, namely, incidence angle and detection distance. These issues may cause discrepancy within the spectral information, thus limiting the classification capabilities of multispectral lidar. To fill this gap, a hue–saturation–intensity (HSI) color space-based method for multispectral lidar classification is proposed in this study. The proposed scheme does not require radiometric calibration, as the HSI color space is robust to spectral intensity variations within a single target. In this method, spectral data are transformed from red–green–blue (RGB) color space to HSI color space. The three components of the HSI color space are inputted for the classification. Then, a reference target-based radiometric calibration is conducted for comparison. The complex indoor scene and the random forest classifier are used for the validation. The classification results of using raw RGB data, raw HSI data, calibrated RGB data, and calibrated HSI data are compared. Results show that the raw HSI data outperform the raw RGB data in terms of classification accuracy. In particular, the raw HSI data can correct the classification error caused by the measurement geometry more effectively than the calibrated RGB data. The improvement resulting from using the HSI color space is demonstrated by both the three-wavelength multispectral lidar and the 32-channel multispectral lidar. That indicates that HSI color space is a promising tool for enhancing the classification capability of multispectral lidar.**

***Index Terms*— Hue–saturation–intensity (HSI) color space, measurement geometry, multispectral lidar, target classification.**

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I. Introduction

SPECTRAL imaging technology reveals information about the physical and chemical attributes of targets. It has been wildly applied in vegetation health management [1], land cover mapping [2], water environment protection [3], and atmospheric detection [4]. Multispectral imaging technology, a typical tool for obtaining abundant spectral information [5], presents the acquired data as 2-D images. Researchers have attempted to fuse data from multispectral imaging and lidar to achieve 3-D spectral imaging [6], [7], but this scheme involves geometric registration problems [8], [9]. Multispectral lidar can address this problem with novel 3-D spectral imaging technology [10]–[12]. The development of multispectral lidar has enabled 3-D spectral imaging to become increasingly practical and effective [13]–[17]. Much research has been conducted on multispectral lidar, particularly the exploration of its potential in earth observation [18]–[21], in which data classification is a fundamental process.

Numerous studies show that the spectral information of multispectral lidar can substantially facilitate data classification. Suomalainen *et al.* [22] performed a classification of Norway spruce by combining a spectral correlation mapper and a region growing algorithm. Chen *et al.* [23] conducted a two-step classification of vegetable and artificial materials using spectral information and neighborhood spatial label-smoothing to eliminate salt and pepper noise. Altmann *et al.* [24] performed depth estimation using a hierarchical Bayesian method and conducted reflectivity-based scene segmentation on different objects composed of clay. Chen *et al.* [25] identified the optimal spectral bands for compositing true-color features, which were subsequently employed in the support vector machine for classification of color card papers. Other researchers also utilized spectral features for airborne multispectral lidar data classification [26]–[28].

The benefit of using spectral information for classification depends on the precise identification of target-scattering characteristics. However, according to the lidar equation, the spectral information of multispectral lidar is affected by instrumental and atmospheric effects, target-scattering characteristics, and measurement geometry [29], [30]. Among them, the target-scattering characteristic factor, which represents the physical and chemical attributes of a target surface, provides fundamental information for target classification. Thus,

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target-scattering characteristic is the key factor needed to be derived. Meanwhile, measurement geometry is regarded the main interference factor obstructing the precise acquisition of target-scattering characteristics for multispectral lidar, resulting in the discrepancy in spectral intensity and limited accuracy of the classification [31], [32]. Many studies try to remove the effect from the measurement geometry (i.e., incidence angle and detection distance). The spectral index is used to calibrate the effects of incidence angles [33]. However, subsequent research has revealed the varying influences of incidence angles on different wavelengths [30]. The spectral index could not calibrate the effects of incidence angle thoroughly. Moreover, reference target-based [34], [35], theoretical model-based [36], [37], and empirical model-based [38] radiometric calibration methods were developed to obtain precise spectral information. Nevertheless, the aforementioned radiometric calibration methods remain to be limited by calibration effects or low applicability for different instruments and scenes [39]–[41].

The hue–saturation–intensity (HSI) color space is introduced in this research for multispectral lidar data classification, as this color space is more robust to intensity variations than the other color spaces [42]. In the HSI color space, the components of hue, saturation, and intensity are extracted individually for color representation, and these three components are relatively independent; by contrast, the three components of the red–green–blue (RGB) color space are highly correlative [43]. In addition, HSI color representation is compatible with the vision psychology of the human eye [44].

Considering these advantages, the HSI color space is applied to alleviate the influence of intensity variations within a single target in the fields of remote sensing and natural image processing. The HSI color space was previously combined with different algorithms in remote-sensing images for cloud detection [45] and land change detection [46]. In the pan-sharpening domain, the HSI color space can separate spectral information from spatial information. Then, the spatial information can be manipulated independently, while maintaining the overall color balance of the original image [47]. In scenes in which the illumination condition changes, the HSI color space has been used extensively to detect road signs [48], human skin [49], trees [50], and the sort of fruit [51]. The HSI color space can also be applied in histogram equalization to avoid undesirable hue shifts during intensity change [52].

The HSI color space was introduced to multispectral lidar data classification in this research owing to its robustness to intensity variations. A complex indoor experimental scene was scanned by a multispectral lidar with the following three wavelengths: 466 (blue), 527 (green), and 628 (red) nm. The raw RGB data, which were normalized by the Spectralon target as the diffuse reflection reference, were initially obtained. Then, the calibrated RGB data were acquired by reference target-based radiometric calibration. The raw and calibrated HSI data were transformed from the raw and calibrated RGB data, respectively. Subsequently, four types of data (i.e., raw RGB, raw HSI, calibrated RGB, and calibrated HSI data) were classified by the random forest classifier. Random forest was selected in this work because it supplies an acceptable tradeoff between accuracy and efficiency [53], and it has been successfully used for remote-sensing data for years [54]. The experimental results were compared to validate the effectiveness of the HSI color space in improving multispectral lidar data classification.

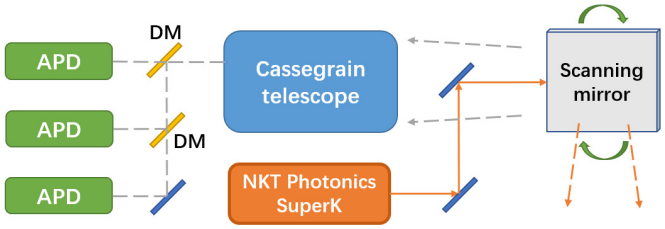


Fig. 1. Brief illustration of the multispectral lidar system.



Fig. 2. Experimental scene that includes 14 targets with different spectral and spatial properties. The serial number in the photograph is in accordance with that in text.

II. Experiment Instruments and Materials

A. Experimental Instruments

The multispectral lidar used in this research was a modified version of the system designed by Song *et al.* [55]. The brief illustration is shown in Fig. 1. The improvement was focused on the change of the laser source; that is, three solid-state lasers at three wavelengths were replaced with a supercontinuum laser (SuperK, NKT Photonics). The pulse repetition rate was 20 kHz, and the beam divergence angle was less than 0.1 mrad. The supercontinuum laser beam was reflected to the targets using a two-axis rotatable mirror for 3-D spatial scanning. Then, the backscatter signal from the targets was received by a Schmidt–Cassegrain telescope. Dichroic mirrors (DM) and optical filters were placed before three avalanche photodiodes (APD) to divide the backscatter signal into three wavelengths, namely, 466 (blue), 527 (green), and 628 (red) nm, for the acquisition of multispectral information. The other details of the system were similar to those in the literature [55].

B. Experimental Materials

The experiment was conducted in an optical laboratory at Wuhan University. As illustrated in Fig. 2, the spatial relationship of experimental materials is complex, which increased classification difficulty. The experimental detection distance was approximately 6.5–7 m from the telescope receiver. This

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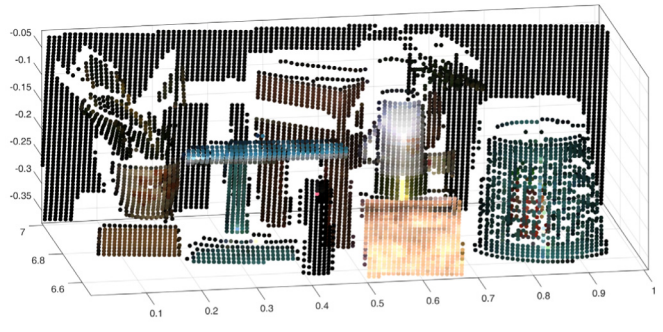


Fig. 3. Multispectral lidar point cloud with raw RGB data.

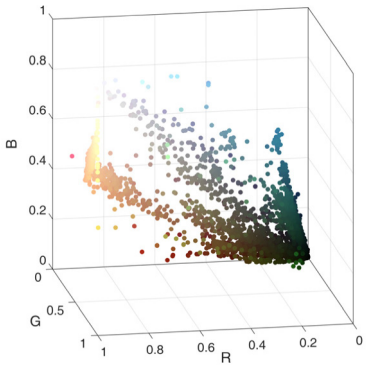


Fig. 4. Distribution of the raw RGB data in the RGB color space. The color of points is consistent with the color in Fig. 3.

value for the experimental distance was selected because of the overlap distance of our system (5 m) and the longest distance in our laboratory (8 m). The 14 classes considered in the experimental scene were as follows: 1) black paper, 2) Sansevieria trifasciata plant, 3) ceramic flowerpot, 4) brown cardboard, 5) blue lamp, 6) black ceramic vacuum cup, 7) white ceramic drinking cup, 8) yellow cellulose tape, 9) wooden, 10) Pachira macrocarpa leaves, 11) Pachira macrocarpa trunk, 12) blue plastic bin, 13) green, and 14) orange part of the carrot-like ceramic object. The materials with different spectral or spatial properties were classified into two classes even if they were spatially connected. For instance, the vegetation and flowerpots and the orange and green parts of the carrot-like ceramic object in the bin were divided into two classes.

The experimental scene consisted of various materials with different spectral and spatial attributes. The materials with relatively high reflectance (e.g., white ceramic drinking cup) and low reflectance (e.g., black paper and black ceramic vacuum cup) were used to validate our model's performance based on the different spectral-featured materials. Moreover, the materials were in different shapes, such as cuboid, cylinder, or vegetation structure. Thus, the validation of our proposed method was based on man-made and natural materials with different spatial and reflective properties in a complex experimental scene.

C. Experimental Data

The point cloud obtained by the multispectral lidar for scanning the experimental scene was used as the experimental data. The multispectral lidar point cloud contained spectral information (intensities of three channels in RGB) and spatial information (coordinates in XYZ). The spectral intensity of each point was normalized by Spectralon targets [56]. The normalized RGB intensities were utilized as the raw RGB data. The point cloud with the raw RGB data is shown in Fig. 3. The size of the point cloud was 1.01 m × 0.51 m × 0.35 m, and 6684 points were obtained.

As shown in Fig. 3, the intensity of the points was considerably affected by detection distance and beam incidence angle. The points at the back were darker than the points at the front because of the influence of detection distance on intensity. The points of the white ceramic drinking cup, cellulose tape, and wooden box were the brightest because of their relatively short detection distance and high reflectance. The peripheral points of the cylindrical materials were darker than the center points because of the influence of the beam incidence angle and detection distance.

The distribution of all points in the raw RGB color space is shown in Fig. 4. Most points tend to be dark, and their RGB values are low. Thus, radiometric calibration needs to be performed on detection distance and beam incidence angle by employing reference target-based radiometric calibration (Section III-B).

III. Methods

The workflow of this research was arranged as follows. The raw RGB data are transformed from the RGB color space to the HSI color space to extract the components of hue, saturation, and intensity (Section III-A). For the comparison part of this study, the raw RGB data are radiometrically calibrated to obtain the calibrated RGB data by employing reference target-based radiometric calibration (Section III-B). Then, the calibrated RGB data are also transformed to calibrated HSI data for comparison, and four types of data (raw RGB, raw HSI, calibrated RGB, and calibrated HSI data) are classified by random forest. The comparative results are used to evaluate the performance of the four data types on multispectral lidar classification.

A. Transformation of RGB Color Space to HSI Color Space

The HSI color space defines numerically the values of the hue, saturation, and intensity of a color [57]. The coordinate axes of the HSI color space are analogously cylindrical, as shown in Fig. 5. One color can be represented as a point with three coordinates (saturation (S), hue (H), intensity (I)) in the HSI color space. Saturation represents the radial distance to the vertical gray-scale axis, while hue is relative to the angle in the polar coordinate perpendicular to the vertical axis. Intensity is the distance to the bottom along the vertical axis.

H is in the range of 0° to 360°; it starts with red at 0°, then green and blue at around 120° and 240°, and ends with red at 360°. H refers to the spectral wavelength of a color. S is in the range of 0–1; it starts at the vertical axis and increases radially and linearly outward to the surface of the cone.

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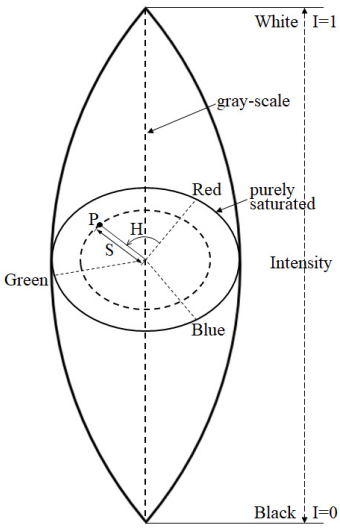


Fig. 5. HSI color space: An approximately cylindrical coordinate.

S measures the degree of dilution of a pure light by the white light. A higher S indicates a richer and gaudier color. I is in the range of 0 at the bottom to 1 at the top of the vertical axis, indicating relative brightness.

At the two ends of the vertical axis are $I = 0$ and 1, referring to colors of absolute white and absolute black, respectively. H and S are undefined and meaningless at these two points. In addition, the S component of all points on the vertical axis is zero, and the H component is undefined. These singularities manifest when the values of R, G, and B are equal.

The following equations are used for the transformation from RGB to HSI [57]:

$$H = \cos^{-1}\left[\frac{0.5[(R-G)+(R-B)]}{\sqrt{(R-G)^2+(R-B)(G-B)}}\right] \quad (1)$$

$$S = 1 - \frac{3}{R+G+B}[\min(R,G,B)] \quad (2)$$

$$I = (R+G+B)/3 \quad (3)$$

where $H = 360° - H$ when $(B/I) > (G/I)$. The ranges of R, G, and B are [0,1] for calculating I in (3). In the classification, H is normalized within the range of [0, 1] by letting $H = H/360°$, and the ranges of S and I are also from 0 to 1. H is undefined when S is zero. Similarly, S is undefined when I is zero.

The HSI color space has advantages over RGB color space because it can directly indicate brightness and chrominance. The component of I is separated from H and S, which means that the relative brightness is disjointed from the chrominance information. Moreover, substantial spectral attributes of the materials can be revealed because the components of H and S are directly related to chrominance information [58]. Therefore, these advantages enable the HSI color space to be an ideal tool for application, especially when chrominance is more significant than luminance.

The advantages of using the HSI color space addresses the situation of intensity variation in multispectral lidar data caused by measurement geometry. In RGB color representation, the discrepancy caused by measurement geometry occurs in all R, G, and B components. By contrast, the discrepancy in HSI color space mainly occurs in the component of I, and the components of H and S remain stable in a single target. Therefore, the HSI color space can likely improve the classification results, especially when the spectral information is interfered by measurement geometry.

However, the HSI color space also has its shortcomings, including singularities during the transformation (incalculable H for achromatic points with equal RGB values) and the sensitivity of H to slight deviations in RGB values for points near the achromatic axis. The influences of these drawbacks on multispectral lidar classification are discussed in subsequent sections.

B. Radiometric Calibration Based on Reference Target

The proposed HSI color space-based method can be validated by comparison with the classification results of calibrated RGB data obtained by reference target-based radiometric calibration. The four key factors affecting the backscattered signal received by a lidar detector include instrumental and atmospheric effects, target-scattering characteristics, and measurement geometry. The following lidar equation describes the effects of these factors on the power of the received signal:

$$P_r = \frac{P_t D_r^2 \rho \cos\theta}{4R^2}\eta_{\text{sys}}\eta_{\text{atm}} \quad (4)$$

where P_r is the received power, P_t is the transmitted power, D_r is the diameter of the receiver aperture, ρ is the reflectance of the target, θ is the incidence angle of the beam, R is the range (detection distance), and η_{sys} and η_{atm} denote the power attenuation caused by the system and atmospheric propagation, respectively. Therefore, in the aspect of measurement geometry, the power of the received signal is proportional to $\cos\theta$ and inversely proportional to R^2.

Considering the short detection distance in our experimental setup, the atmospheric attenuation on the received signal is not discussed in this study. Instead, the radiometric calibration is focused on the influence of measurement geometry, including detection distance and incidence angle. The reference target-based method is used in our research because of its desirable calibration performance [35].

The standard reference target Spectralon, with 99% reflectivity, is used to investigate the relationship between multispectral lidar intensity and detection distance or incidence angle. Spectralon is a fluoropolymer with high and stable diffuse reflectance over the ultraviolet, visible, and near-infrared regions of the spectrum [56].

Reference target-based calibration is composed of two parts, namely, the calibration of detection distance and the calibration of incidence angle. For the detection distance, the Spectralon target is placed at varying distances ranging from 6.5 to 7 m, covering all points of the experimental scene. The Spectralon target is set perpendicular to the receiver of the multispectral lidar. As for the incidence angle, the Spectralon target is placed at varying incidence angles ranging from 0° to 70° at a distance of 6.5 m. The distance of 6.5 m is selected because the experimental distance is longer than the overlap distance,

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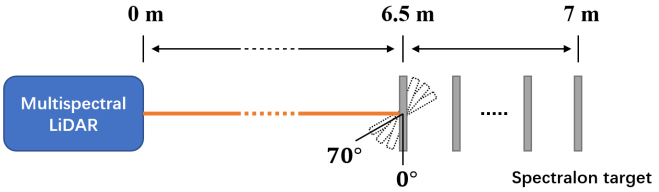


Fig. 6. Reference target-based radiometric calibration experimental setup.

and 6.5 m is the nearest experimental distance with the highest signal to noise ratio. The radiometric calibration experimental setup is illustrated in Fig. 6.

The intensity of the Spectralon target in each setup was obtained thrice by the multispectral lidar, and the average value was adopted. These intensities were normalized at a distance of 6.5 m and an incidence angle of 0°. Linear interpolation was conducted to achieve the curves of the relationships between normalized intensity and detection distance or incidence angle. The relationship curves are shown in Fig. 9 in Section IV-B. Then, the intensity of the raw RGB point cloud was calibrated to a detection distance intensity of 6.5 m and an incidence angle of 0° based on the relationship curves. The distance was directly taken from the multispectral lidar, and the incidence angle was calculated using the CloudCompare ver. 2.10 software. The intensity of the Spectralon target with an incidence angle larger than 70° is difficult to measure. A large incidence angle may also induce an "over-correction" effect [59], [60]. Therefore, the intensity of points with incidence angles larger than 70° in this work were calibrated as if their incidence angles were 70°. The threshold of 70° could be larger than the common threshold of airborne lidar [61], [62].

C. Classification and Evaluation

Random forest is often used as a classifier because it offers an acceptable tradeoff between accuracy and efficiency [53], and it can handle numerous input variables and has been used for years in lidar point cloud classification [54].

The random forest is a typical ensemble learning method, and it can be used for both classification and regression. A random forest combines a collection of decision trees and adds additional randomness while growing the trees to produce a more robust result. A decision tree is a tree-like model producing possible outcomes with a collection of associated choices. A decision tree structure contains internal nodes, branches (directed edge), and leaf (terminal) nodes. Each internal node conducts a repartition based on one feature. Each branch of an internal node represents a consequence of the repartition, and a leaf node indicates a final class label. This growing rule leads to a tree-like form.

In random forest, each tree is trained by a random subset of the training samples and features, and the tree produces its own response for the outcome label. Then, the most popular class voted by the ensemble of decision trees is adopted as the final result. Different from the single decision tree, random forest with tree ensembles can remarkably improve robustness and classification accuracy of a data set.

In this research, the classification feature sets used were simply the 3-D vectors of RGB or HSI. The range or geometry was not included because this research focused on the performance of the HSI color space. For the configuration of the classification, the random forest contained 20 trees to ensure robustness. Due to the small data set, the fivefold cross validation [63] was used for training and evaluation as a means of avoiding overfitting. The data used in the fivefold cross validation were first divided into five disjoint folds. Then, a model for each fold was trained using the out-of-fold observations (training points), and the model's performance was assessed using in-fold data (checking points). The results of the five in-fold data were combined as the final classification result, and quantitative evaluation was subsequently conducted accordingly.

Overall accuracy (OA), precision, and recall were introduced into the quantitative evaluation. The OA was used for the overall evaluation, while precision and recall were used for the single-class evaluation. Their formulas are as follows:

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{5}$$

$$\mathrm{precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{6}$$

$$\mathrm{recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{7}$$

$$p_e = \frac{\sum_{j=1}^{J} a_j \times b_j}{n^2} \tag{8}$$

where TP is true positive, TN is true negative, FP is false positive, and FN is false negative. a_j and b_j indicate the true and predicted numbers of a sample in class $j \in J$, and n is the total number of samples.

IV. Results

A. Result of Transformation From RGB to HSI Color Space

This section describes the results of transformation of the raw HSI data from the raw RGB data. Fig. 7(a)–(c) show the normalized H, S, and I values of the point cloud in grayscale ranging from 0 to 1. H provides a measure of the spectral wavelength of a color. The H values of red, yellow, green, and blue are 0, 0.15, 0.3, and 0.5, respectively. Therefore, the points of the red, yellow, and green materials in Fig. 7(a) are relative dark, while the points of the blue plastic lamp and bin are gray. The H values of the chromatic materials are stable. However, for achromatic (black, gray, and white) materials, their H values are unstable because of the sensitivity of H to slight deviations of RGB values near the grayscale points. Therefore, considerable variations exist on the H values of the black ceramic vacuum cup and the white ceramic drinking cup, as shown in Fig. 7(a). The variation of H at the upper part of the black paper may be caused by the radiometric instability of the multispectral lidar. A hot spot area appears on the white ceramic drinking cup; this hot spot may be caused by the incidence angle and the ceramic material. The hot spot not only results in the variation of I but also aggravates the variation of H. In addition, no two points have the same RGB values in our experimental data; thus, no point without H is observed.

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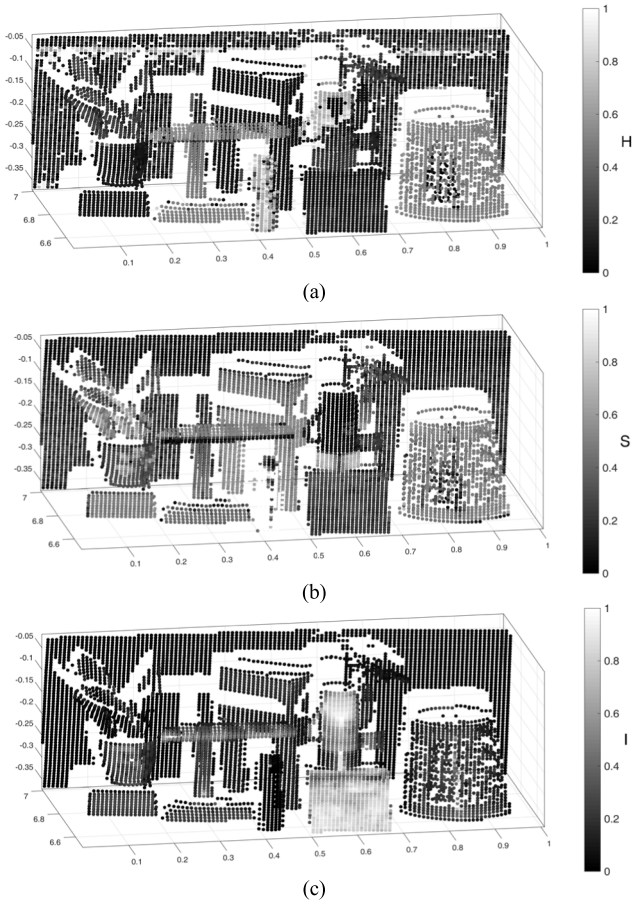


Fig. 7. Raw HSI data: normalized (a) H, (b) S, and (c) I components in gray scale transformed from the raw RGB data.

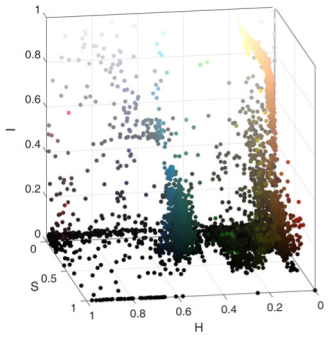


Fig. 8. Distribution of the raw HSI data in HSI color space. The color of points is the raw RGB color obtained by multispectral lidar, which is consistent with those in Figs. 3 and 4. The points in the same target tend to share similar H and S components.

Saturation measures the degree of dilution of pure light by white light. In general, the richer the color is, the larger the value of S becomes. Equation (2) indicates sensitivity to slight deviations in RGB values on very dark materials, as shown on the black ceramic vacuum cup in Fig. 7(b). Meanwhile, as shown in Fig. 7(c), intensity measures relative brightness. The point cloud with raw RGB color is shown in the normalized HSI color space in Fig. 8.

Colors with different H, S, and I values can be separated, as shown in the HSI color space in Fig. 8. The points on the wooden box are yellow or orange, with H ranging from 0.05 to 0.18. These areas are also relatively bright, with the I ranging from 0.5 to 1. The vegetation points are dark green, with H ranging from 0.2 to 0.4 and I lower than 0.2. The relatively low intensity of vegetation can be attributed to low reflectance. The points of the blue plastic lamp and bin are closely distributed, with H ranging from 0.4 to 0.6 and I ranging from 0.05 to 0.6.

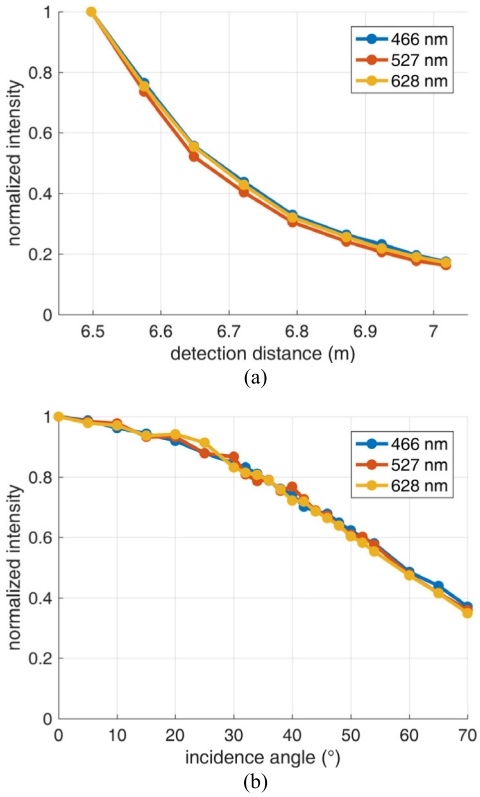


Fig. 9. Relationship curves between the normalized intensity and (a) detection distance, and (b) incidence angle of three wavelengths.

B. Result of Reference Target-Based Radiometric Calibration

After the reference target-based radiometric calibration, the normalized intensity was obtained under different detection distances and incidence angles for the three channels. These intensities were normalized at a distance of 6.5 m and an incidence angle of 0°. The curves of the relationship between normalized intensity and detection distance or incidence angle are shown in Fig. 9. The curves indicate a decline in normalized intensity with the decrease in detection distance and incidence angle. The detection distance curve is similar to the quadratic function. By contrast, the incidence angle curve is similar to the cosine function, and it basically accords with the relationship among received power, detection distance, and incidence angle in the lidar equation.

The curves were obtained through linear interpolation. The raw RGB data were normalized in terms of intensity at a distance of 6.5 m and an incidence angle of 0° and according to the curves for obtaining the calibrated RGB data. Fig. 10 shows the calibrated RGB point cloud, and

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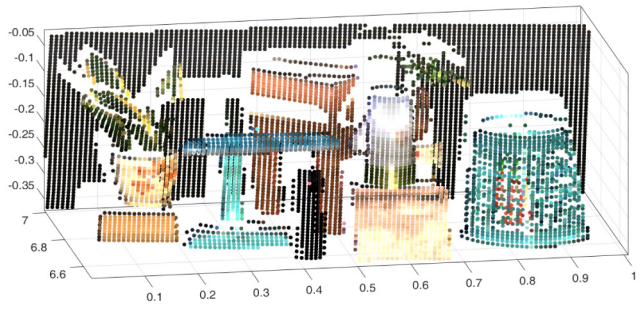


Fig. 10. Multispectral lidar point cloud with the calibrated RGB data after reference target-based radiometric calibration.

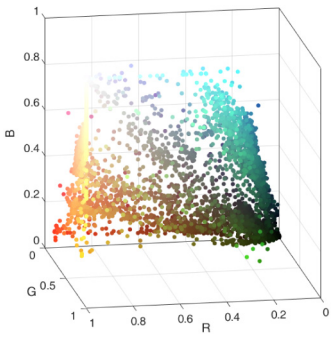


Fig. 11. Multispectral lidar point cloud's distribution in the RGB color space after reference target-based radiometric calibration. The color of points is consistent with the color in Fig. 10.

Fig. 11 shows the calibrated RGB point cloud's distribution in the RGB color space.

Most points obtained higher RGB values after the reference target-based radiometric calibration. The color of the point cloud became brighter. The materials with brighter colors in Fig. 10 are easier for humans to distinguish than those in Fig. 3. In addition, as shown in Fig. 11, the distribution of the calibrated RGB data is more scattered than that of the raw RGB data shown in Fig. 4.

Subsequently, the calibrated RGB data were transformed into calibrated HSI data. Fig. 12(a)–(c) shows the normalized H, S, and I values of the calibrated HSI data in grayscale in the range of 0–1. The distribution of the calibrated HSI data in the normalized HSI color space is shown in Fig. 13.

A comparison between Figs. 7 and 12 indicates that the H and S components of the calibrated HSI data are similar to those of the raw HSI data. The main difference is in the I components, as the influence of radiometric calibration was chiefly on the I component. The influence on the I component can also be derived by comparing Figs. 8 and 13. The I values of most points in Fig. 13 are higher than those in Fig. 8.

C. Classification Results

The four types of data (raw RGB, calibrated RGB, raw HSI, and calibrated HSI data) were classified by random forest. Then, the classification results were compared. Fig. 14 shows the experimental results. Table I shows the quantitative evaluation for the four cases.

The improvements on OA indicate that the HSI color space is more effective than the radiometric calibration

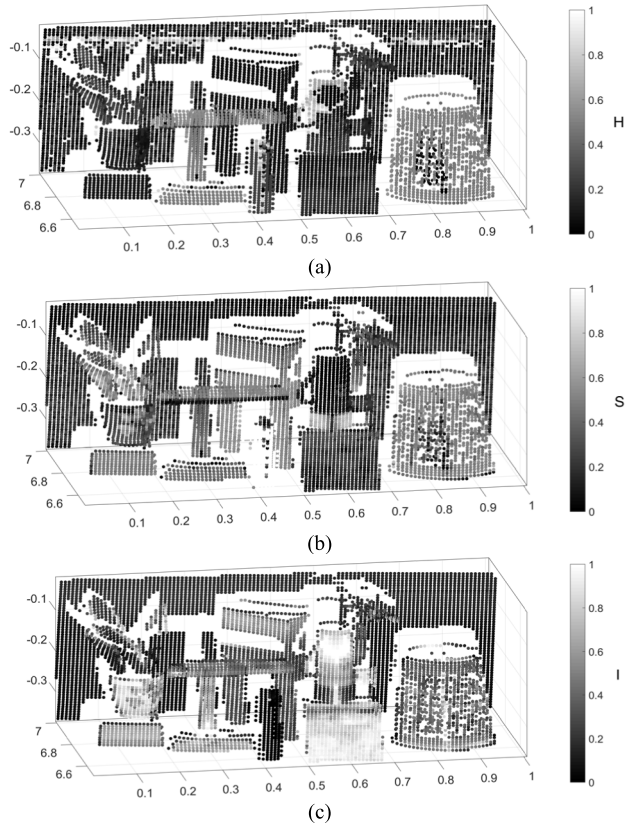


Fig. 12. Calibrated HSI data: normalized (a) H, (b) S, and (c) I components in gray scale transformed from the raw RGB data.

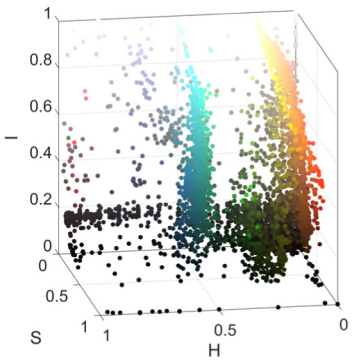


Fig. 13. Distribution of calibrated HSI data in the normalized HSI color space. The color of points is consistent with the color in Figs. 10 and 11.

for multispectral lidar classification. As shown in Fig. 14, the results of the raw and calibrated HSI were quite similar. The raw HSI data obtained the highest accuracy among the four data types. The raw HSI data increased by 7.00% on OA relative to the raw RGB data. Although the value of the increase was not significant, the improvement on the classification error caused by the measurement geometry can be observed (Fig. 14).

The errors on the marginal parts of the blue plastic bin were clearly alleviated in the raw HSI results. Most falsely classified points of the blue plastic bin in the raw RGB results were distributed at the marginal parts [Fig. 14(a) and (b)]. The reason is that incidence angle and distance both change

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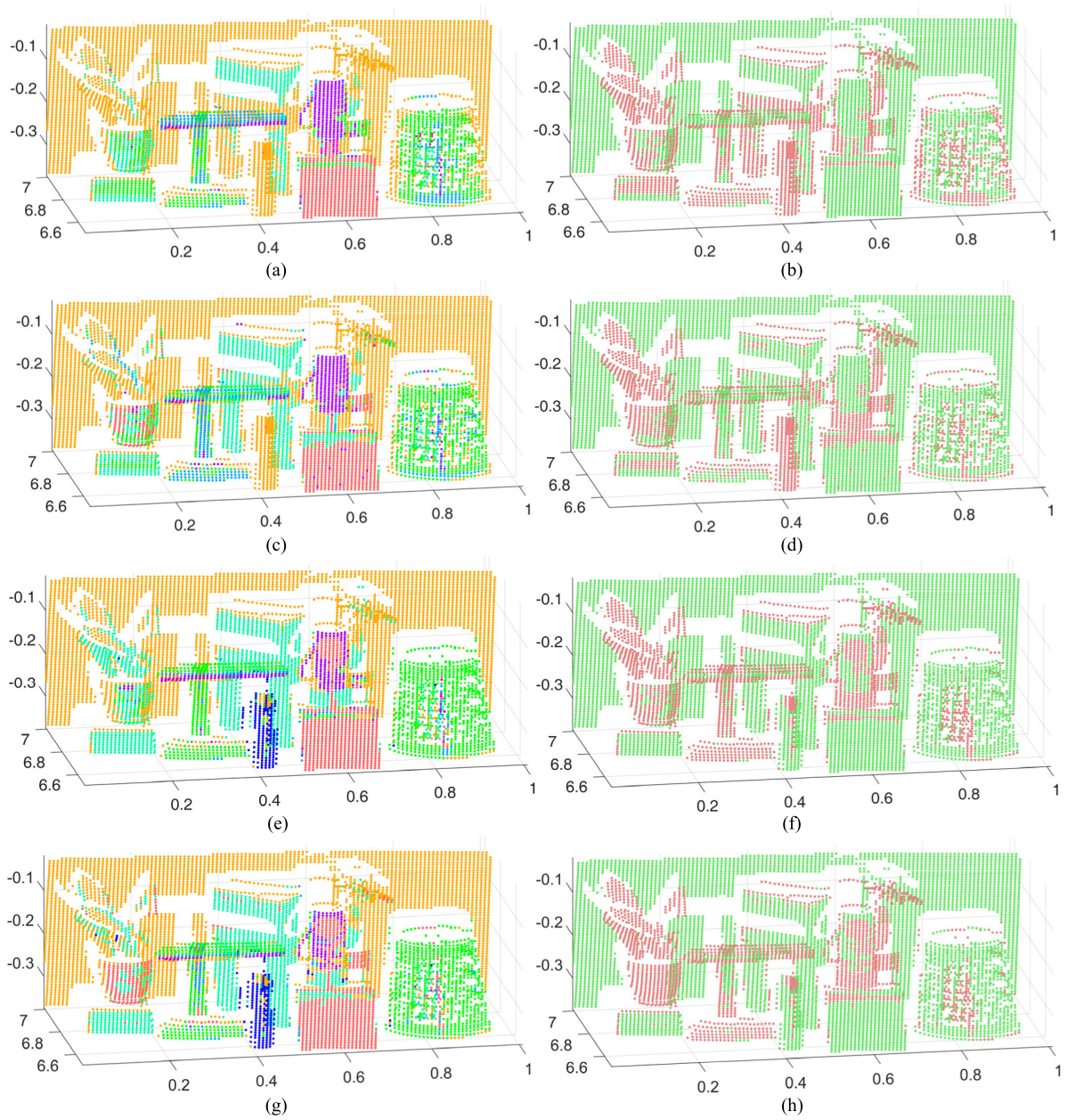


Fig. 14. Classification results based on (a), (b) the raw RGB; (c), (d) calibrated RGB; (e), (f) HSI data; and (g), (h) calibrated HSI data. The left column indicates the various materials with different colors, and the right column indicates the correctly and falsely classified points with green and red colors.

gradually at the marginal parts, and the spectral intensity of these points changes accordingly. These scenarios lead to darker points at the marginal parts and brighter points at the center, as illustrated in Fig. 3. The recall of the blue plastic bin in the raw RGB result was only 0.57. However, the raw HSI data effectively improved the recall to 0.88, which was much higher than 0.71 in the calibrated RGB result. In the raw and calibrated HSI results, the misclassifications of these marginal points were corrected [Fig. 14(f) and (h)], validating the correction effect of the HSI color space on the classification error caused by measurement geometry.

In addition, the HSI color space performed better on the brown cardboard boxes and the black ceramic vacuum cup. The intensity variation also manifested in the brown cardboard box, which could be observed in the calibrated RGB data (Fig. 10). This intensity variation might not be caused by measurement geometry, but also obstruct the classification [Fig. 14(a) and (b)]. Furthermore, the raw HSI data successfully improved its recall from 0.37 to 0.84. The black ceramic vacuum cup in the raw RGB result was totally misclassified as black paper, whereas it was successfully classified with a recall of 0.78 in the raw HSI result because of the difference between its distributions in the raw RGB and HSI color spaces. These findings will be analyzed in Section V.

The raw and calibrated HSI data also generated unsatisfactory results. The blue lamp was classified partially correctly in the raw and calibrated RGB results, whereas it was mostly misclassified as a blue plastic bin in the raw HSI result.

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TABLE I
QUANTITATIVE EVALUATION FOR FOUR CLASSIFICATION CASES

	Raw RGB		Calibrated RGB		Raw HSI		Calibrated HSI	
OA	65.27%		70.33%		72.27%		71.69%	
Class No.	R	P	R	P	R	P	R	P
1	**1**	0.63	0.99	0.71	0.99	0.79	0.99	**0.80**
2	0	0	0	0	0	0	0	0
3	**0.33**	0.42	0.20	0.38	0.16	**0.59**	0	0
4	0.37	0.63	0.69	**0.75**	**0.84**	0.62	**0.84**	0.68
5	0.35	0.61	**0.47**	**0.69**	0.09	0.48	0.16	0.58
6	0	0	0	0	0.78	**0.79**	**0.84**	0.76
7	0.70	0.72	**0.72**	**0.73**	0.44	0.57	0.33	0.72
8	**0.26**	**0.59**	0	0	0	0	0	0
9	**0.88**	**0.96**	0.77	0.72	0.87	0.80	0.85	0.54
10	0	0	0	0	0	0	0	0
11	0	0	0	0	0	0	0	0
12	0.57	0.69	0.71	**0.75**	**0.88**	0.65	**0.88**	0.66
13	**0.44**	**0.52**	0.42	0.29	0	0	0	0
14	0	0	**0.33**	**0.14**	0	0	0	0

R and P represent the recall and precision. The class number on left is in accordance with that in Fig. 2. The highest recall and precision for each class were set to bold.

This finding may be due to the similarity on materials and spectral properties of the two objects. The recall of the white ceramic drinking cup decreased from 0.70 to 0.44 in the raw HSI result. The reason may be that the hue component is sensitive to achromatic materials. The scattered distribution of the achromatic material on the hue component may interfere with the classification, as shown in Fig. 15(c).

Apart from the classes mentioned above, other classes were also poorly classified in both RGB and HSI results. The Sansevieria trifasciata plant was mostly misclassified as black paper in the RGB results because of its low reflectance. Some parts were misclassified as the brown cardboard in the HSI results because of the similar H distributions. In addition, the Pachira macrocarpa leaves were all misclassified because of the low reflectance and spectral variation caused by the complex vegetation structure with many small leaves. The Pachira macrocarpa trunk was partially misclassified as the brown cardboard because of their similar H and S distributions. The green and orange carrot-like ceramic objects in the bin were also poorly classified because of the multi-echo and the small sample data set.

In summary, the classification results of the raw and calibrated HSI data were similar. Furthermore, their OAs were higher than those of the raw and calibrated RGB data. In terms of the correction on the error caused by measurement geometry, the HSI data were more effective than the calibrated RGB data.

V. DISCUSSION

A. Comparison Between the RGB and HSI Color Space

The difference in the classification performance between the RGB and HSI color spaces is attributable to their differences in color space representation. The components of

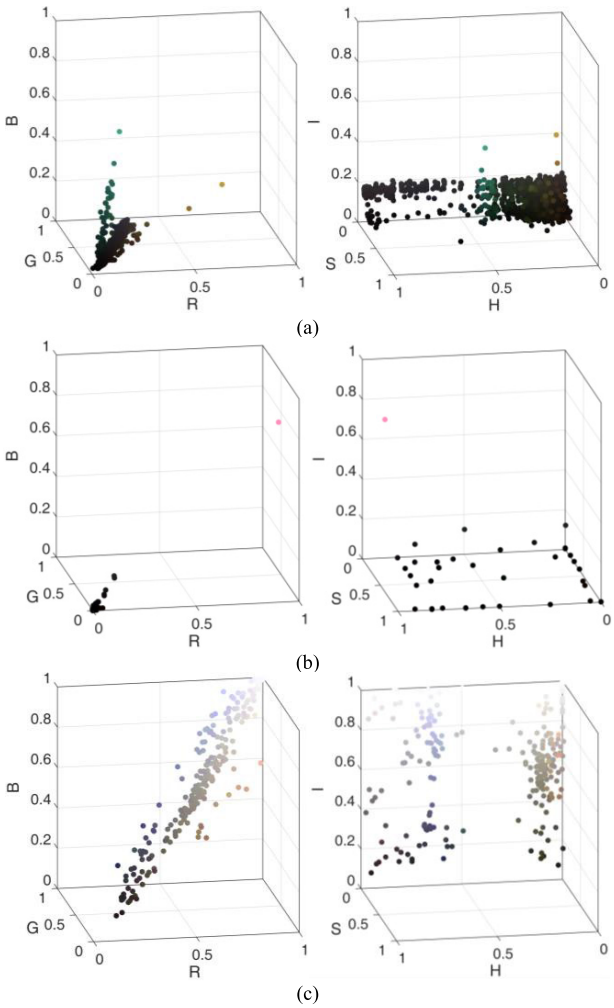


Fig. 15. Point distribution in the (left row) raw RGB and (right row) raw HSI color spaces for (a) the black paper, (b) black ceramic vacuum cup, and (c) white ceramic drinking cup.

the RGB representation are highly correlative [43]. By contrast, the three components of the HSI model are relatively independent. The separation of hue, saturation, and intensity can improve classification performance. Three approximately achromatic targets, namely, the black paper, the black ceramic vacuum cup, and the white ceramic drinking cup, can show the distinction between the RGB and HSI color spaces on target representation (Fig. 15).

The distribution of the black paper points in the raw RGB color space is centralized from (0,0,0) to (0.2,0.2,0.2). A small number of points tend to be blue or earth-yellow due to the multi-echo. Moreover, some reflective signals are backscattered from the blue plastic bin and the wooden box. Meanwhile, the distribution in the raw HSI color space is centralized for the I and S components but not for the H component. H ranges from 0 to 1. As the black paper is an achromatic material, its H component is significantly affected by the slight variations in the RGB values. The wide distribution of the H component is also shown in the black ceramic vacuum cup and the white ceramic drinking cup [Fig. 15(b) and (c)].

The distribution of the black ceramic vacuum cup in the raw RGB color space is more concentrated than that of the black

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paper and focused on (0,0,0). Only one pink point is far from the main part. This pink point may be caused by the specular reflection of the ceramic surface and the hot spot. Meanwhile, the points in the raw HSI color space are distributed on the bottom plane of HS, except for the pink point.

The points of the white ceramic drinking cup are distributed along the main diagonal from (0.1,0.1,0.1) to (1,1,1) in the raw RGB color space. The points at the center of the cup tend to be bright, whereas the periphery points tend to be dark. Furthermore, some points are pale yellow and pale blue, which are more separable in the raw HSI color space.

B. Feasibility of HSI Color-Based Method for Other Multispectral Lidar Systems

The proposed HSI color-based method is suitable for our multispectral lidar with the three wavelengths of 466, 527, and 628 nm. These three wavelengths are from blue, green, and red bands. The feasibility of using the proposed method for multispectral lidar with other wavelengths will also be discussed.

Even if the infrared wavelength is involved in the multispectral lidar with three wavelengths (w_1, w_2, w_3) instead of the RGB channels, the HSI color-based method is still expected to be feasible. In this situation, w_1, w_2, w_3 can be considered the channel of R, G, and B. The values of H', S', and I' are calculated by (1), (2), and (3), respectively. H', S', and I' are not exactly the conception of hue, saturation, and intensity but the intensity variation caused by measurement geometry, which also mainly occurs in the component of I'. Therefore, we suppose that the HSI color space-based method will also be feasible for other three wavelengths.

As for the multispectral lidar with more wavelengths, the multispectral information can be transformed initially into RGB data and then into HSI data. The multispectral information is initially transformed into chromaticity values according to the tristimulus values of corresponding spectral bands as defined in the International Commission on Illumination (CIE) 1931 XYZ color space [64]. Then, the chromaticity values are transformed into RGB values according to a conversion matrix following the standard illuminants of D65 [65]. Subsequently, the HSI values can be obtained according to (1), (2), and (3). This workflow is technically sound but has drawbacks, as the partial spectral information may be discarded during the transformation. This drawback may further degrade the performance of the classification.

Apart from the three-wavelength lidar used in this research, another 32-channel multispectral lidar covering the range of 431 of 751 nm [25] is considered. The HSI color space method is demonstrated on this 32-channel multispectral lidar to validate the feasibility on multispectral lidar with more wavelengths. The transformation from multispectral data into HSI data is realized by the method described previously. The experimental scene contained seven classes (Fig. 16).

Six types of data are used for classification of the 32-channel multispectral lidar: raw 32-dimensional multispectral data, raw RGB data, raw HSI data, calibrated 32-dimensional multispectral data, calibrated RGB data, and calibrated HSI data.

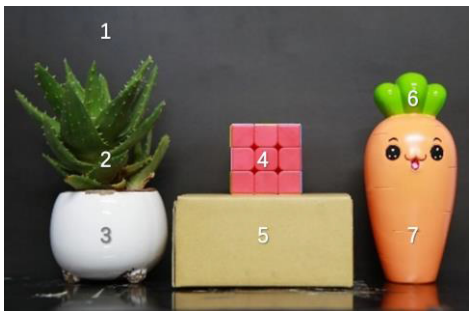


Fig. 16. Experimental scene with seven classes for 32-channel multispectral lidar.

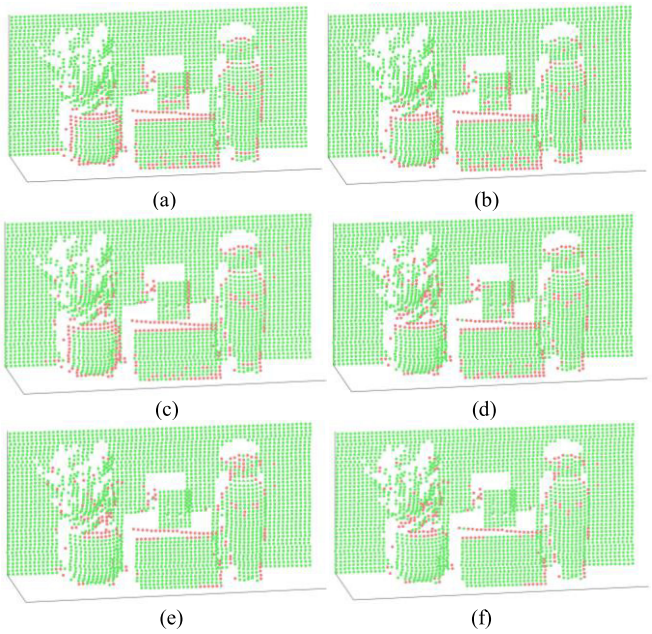


Fig. 17. Classification results based on (a) raw and (b) calibrated 32-dimensional multispectral data; (c) raw and (d) calibrated RGB data; (e) raw and (f) calibrated HSI data. The green and red points indicate the correctly and falsely classified points.

TABLE II
CLASSIFICATION RESULTS OF 32-CHANNEL MULTISPECTRAL LIDAR

	Raw 32	Cal. 32	Raw RGB	Cal. RGB	Raw HSI	Cal. HSI
OA	89.71%	90.43%	90.21%	90.71%	93.07%	93.39%

Cal. indicates the radiometric calibrated data.

The random forest with same configuration mentioned above is used. Fig. 17 and Table II show the classification results.

The calibrated HSI data achieved the highest accuracy and corrected most of the falsely classified points on the marginal parts of the left white ceramic flowerpot, which were caused by the measurement geometry. In addition, the HSI results also performed effectively on the brown cardboard box and the red Rubik's cube. Notably, the raw and calibrated 32-dimentional data obtained slightly worse results than the raw and calibrated RGB data. This finding may be due to the Hughes phenomenon with high dimensional data and limited

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training samples [66]. In summary, the HSI color space is a promising technique for the 32-channel multispectral lidar.

VI. Conclusion

Multispectral lidar is an effective tool for earth observation with the capability of obtaining spectral and 3-D spatial information simultaneously. However, the discrepancy within the spectral information caused by measurement geometry limits its classification and further application. The performance of multispectral lidar classification needs to be improved. Hence, the HSI color space-based method is introduced in this study because of its robustness to intensity variation.

The improvement from HSI color space was demonstrated by both the three-wavelength multispectral lidar and the 32-channel multispectral lidar. The experimental results indicate similar performances of the raw and calibrated HSI data, and using HSI data can remarkably improve accuracy compared with the raw RGB data. The raw HSI data also perform more effectively on the points affected by spectral intensity variation when compared with the calibrated RGB data. These results imply that the HSI color space-based method is an effective tool for correcting classification errors caused by measurement geometry, and it can improve the classification capability of multispectral lidar.

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